

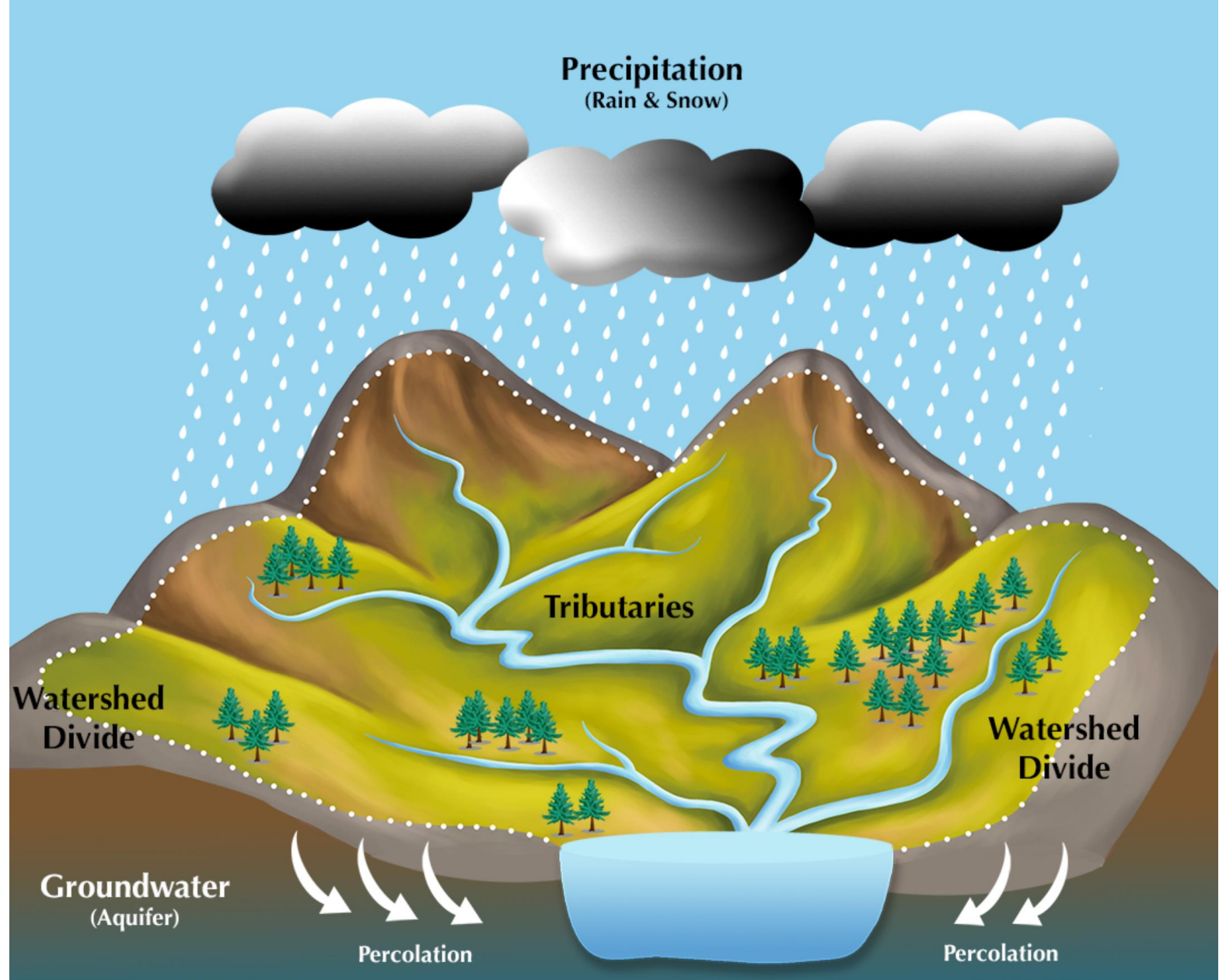
Lab 5: Watersheds

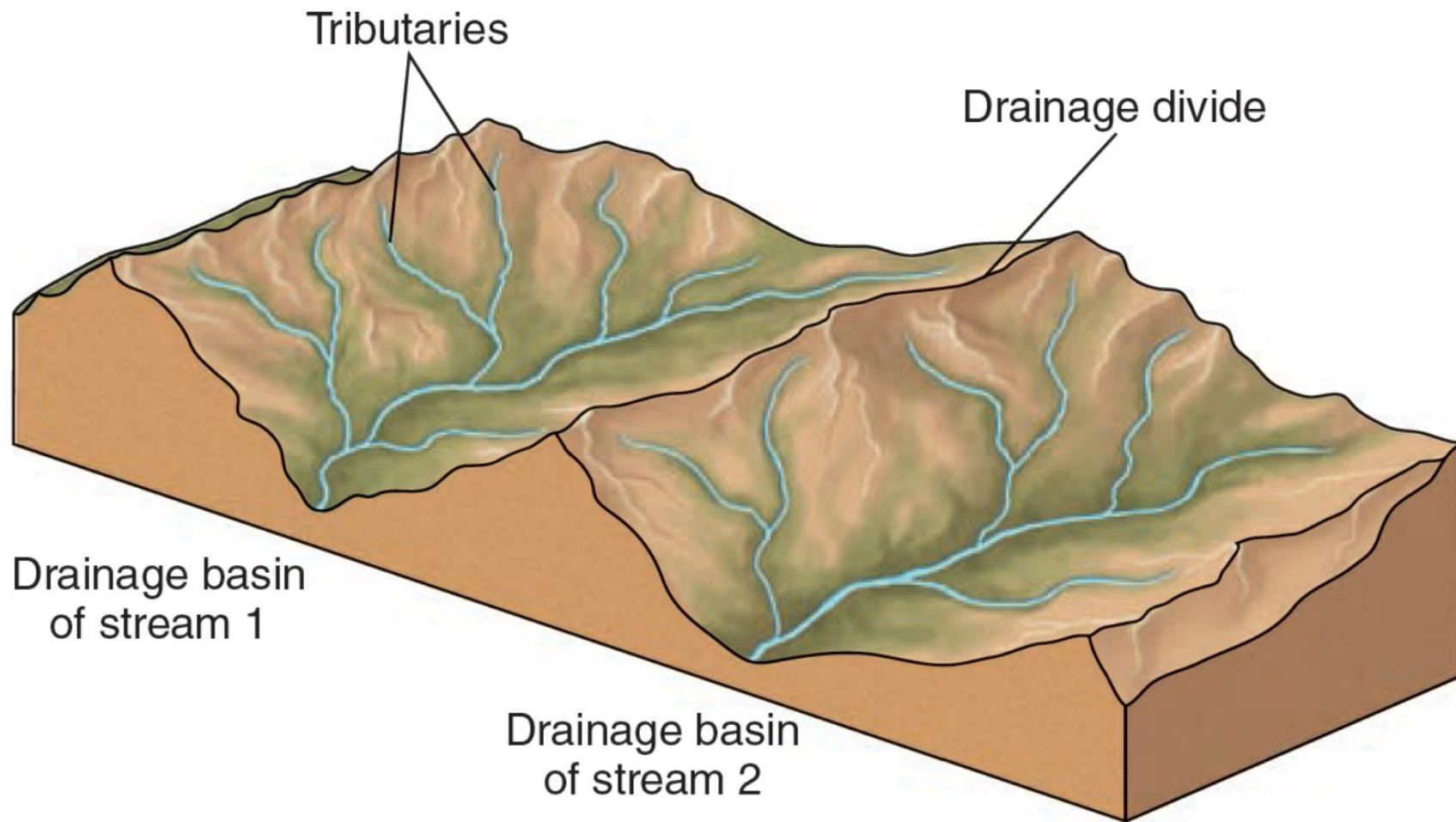
Watersheds

Define the area of land that contributes water to a stream/river

- Drainage basin

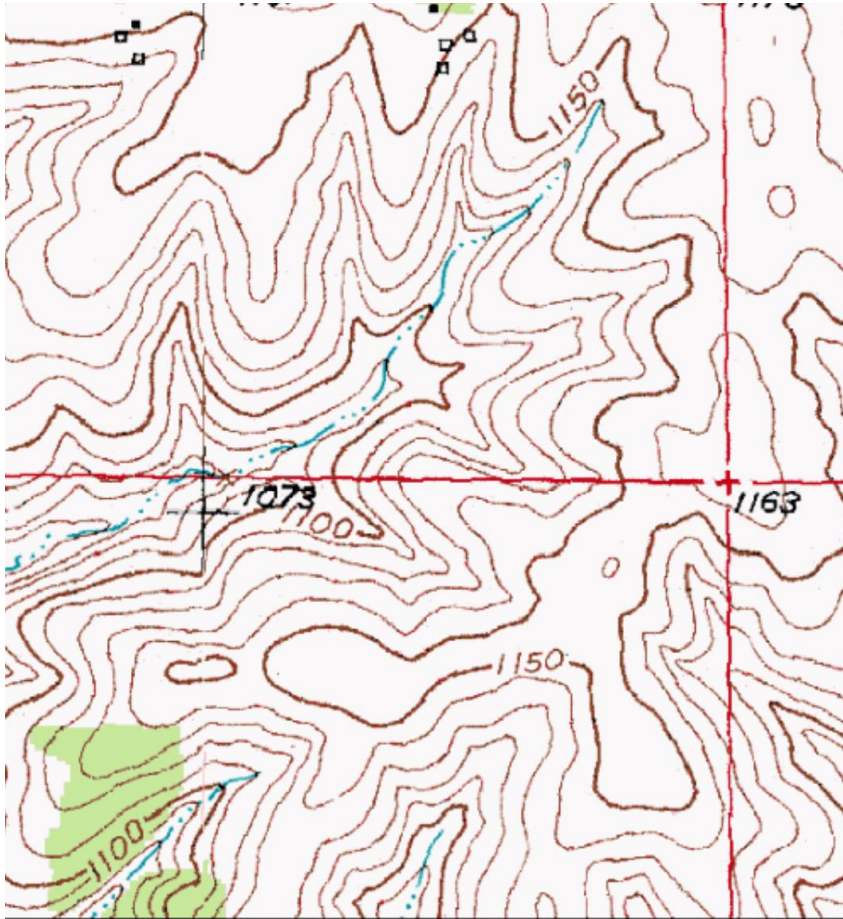
Identifies highest points surrounding a river



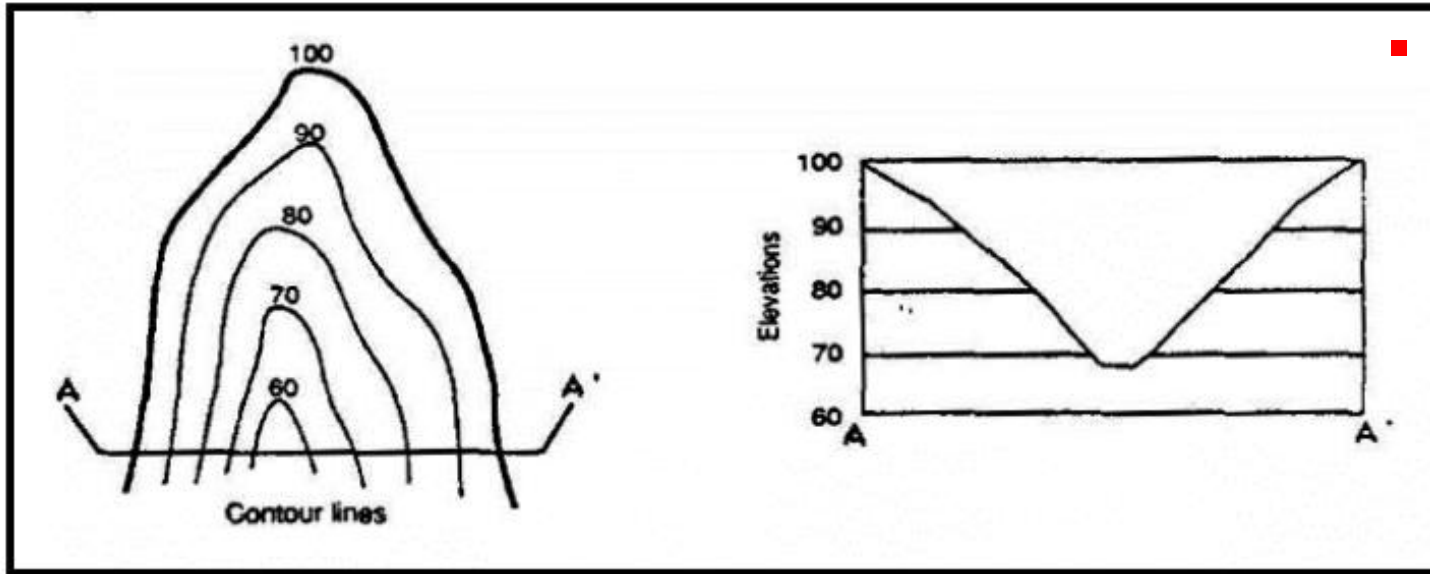


Watersheds

- Topographic maps can help us delineate watershed boundaries
- Review from lab 1:



- Contour lines- lines connecting points of same elevation on the surface
- Index contour – bold lines (every 5th line)
- Contour interval



- Valleys = V's point uphill (towards high elevation)

Figure D-5: Valley

- Ridges = V's point downhill (towards lower elevation)

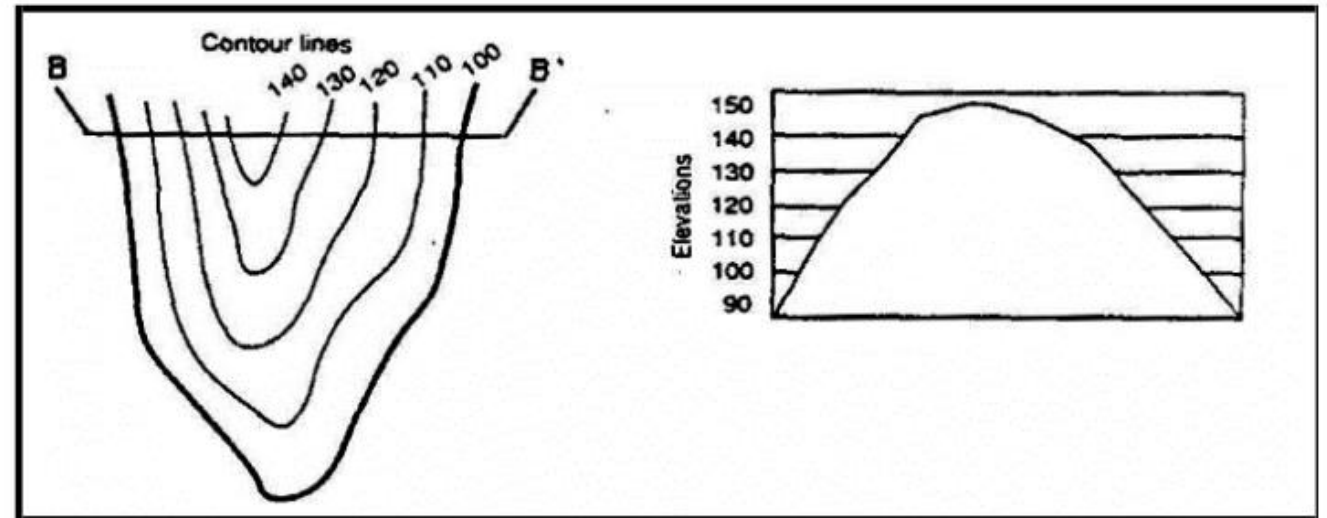
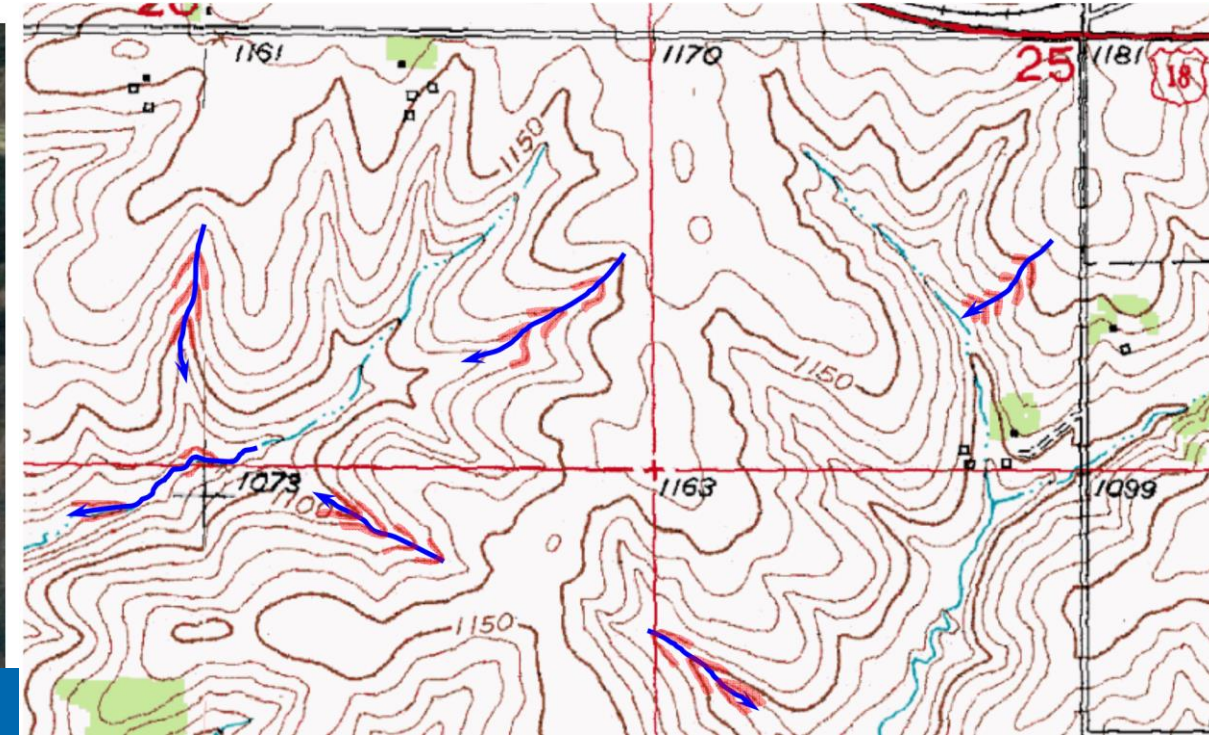
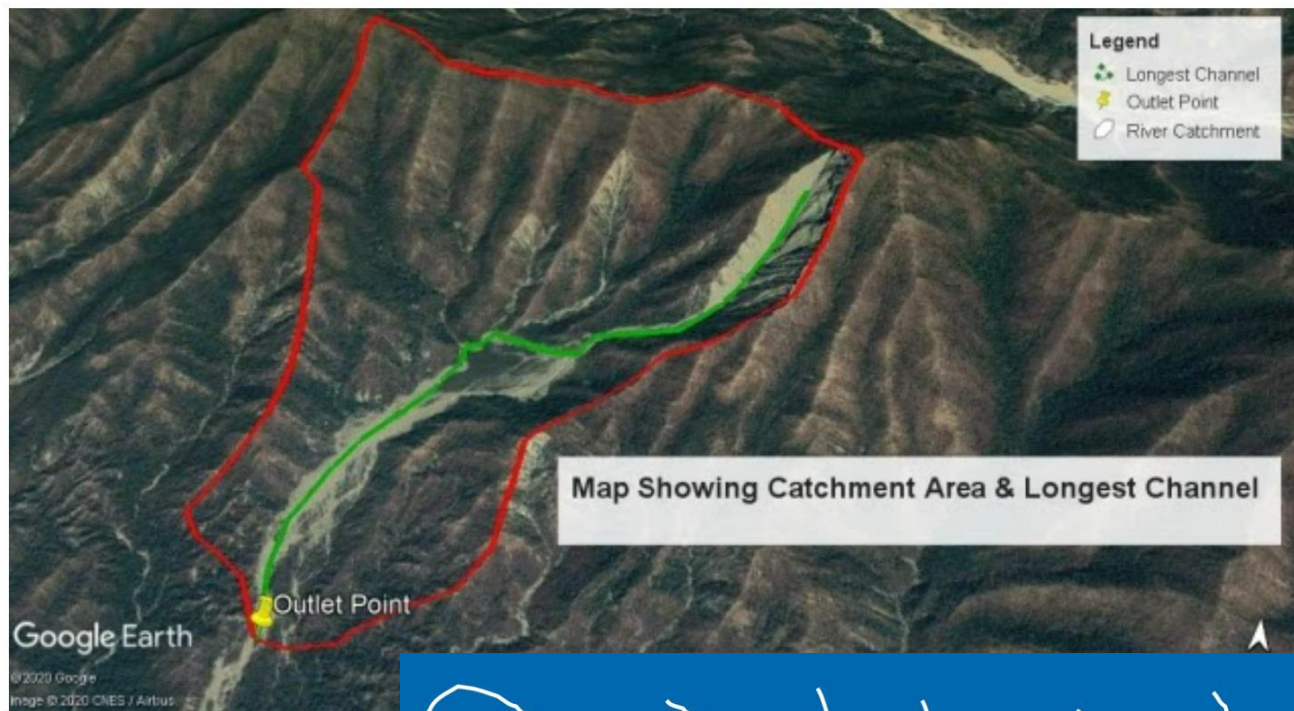
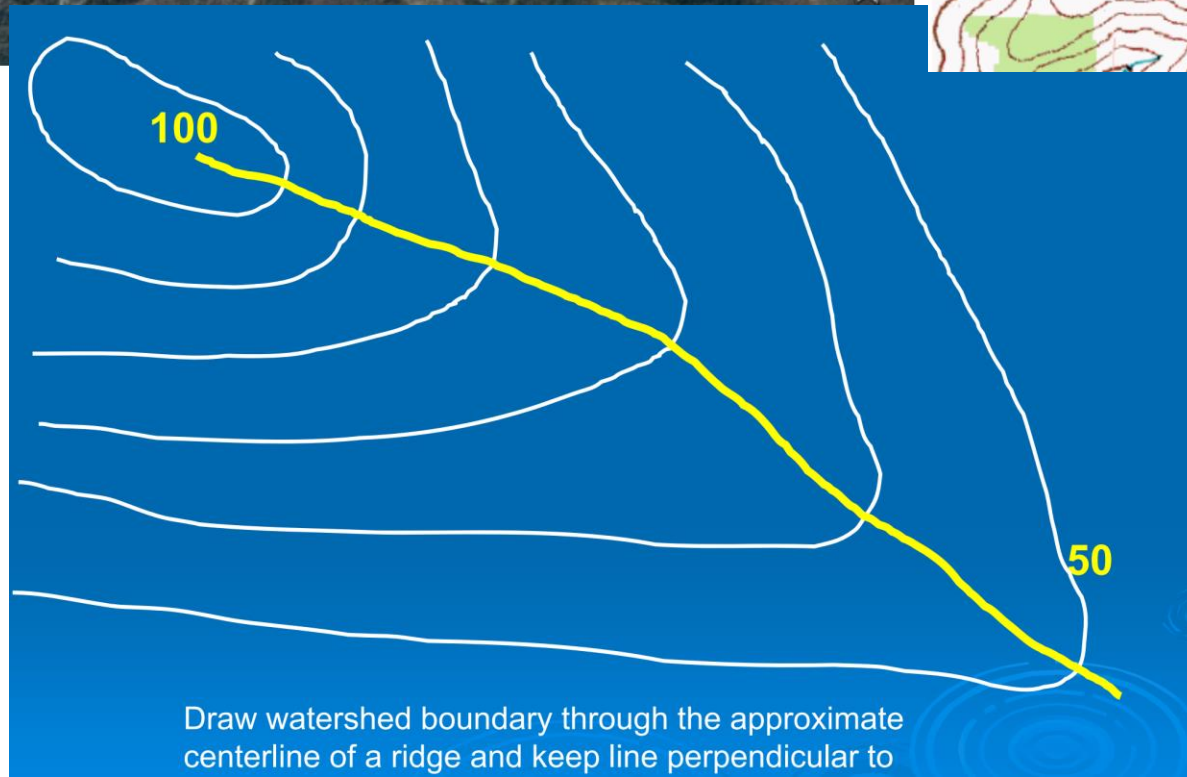


Figure D-6: Ridge



- Ridges = V's point downhill (towards lower elevation)

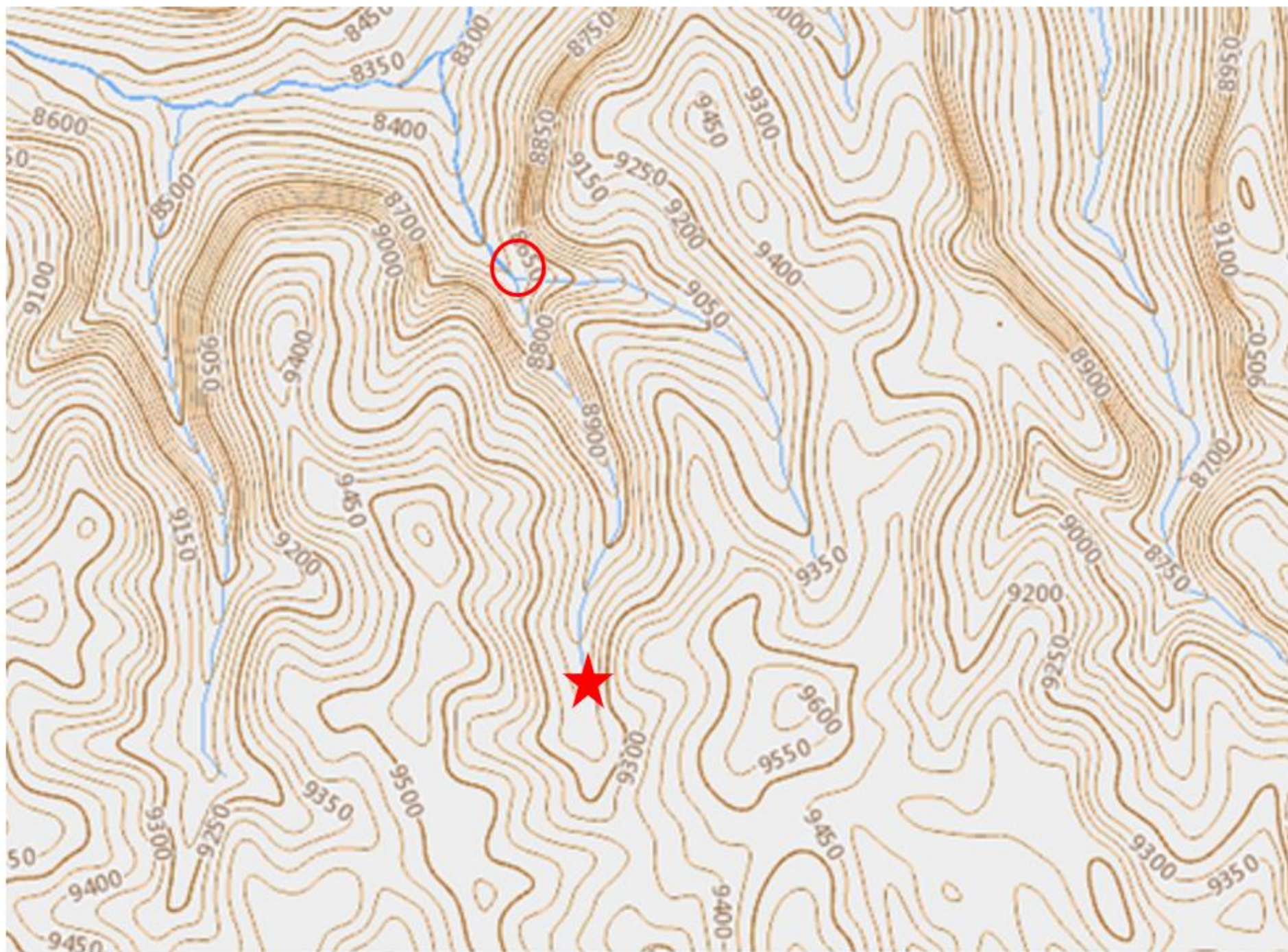


- Valleys = V's point uphill (towards high elevation)

Steps for watershed delineation:

- Step 1: **identify the outlet** and headwater (already labelled in worksheet) **next you may want to roughly mark the high elevation (headwater) and low elevation (outlet) region
- Step 2: **identify the high areas** (look for closed area/circle) that are close to the river
- Step 3: identify valleys and ridges
 - Valleys = V's point uphill (towards high elevation)
 - **Ridges** = V's point downhill (towards lower elevation)
- Step 4: draw the streams along the valleys (no need to do for now)
- Step 5: **connect the high points and ridges with the outlet** with perpendicular line

*****Watershed delineation is perpendicular to contour lines*****



Water budget

A simple comparison of water supply and atmospheric demand

$$\text{Water budget} = \text{Precipitation (P)} - \text{Potential evapotranspiration (PE)}$$

Precipitation (P)

Water coming in
from rain or snow

compared with



Potential ET (PE)

How much water the air
could remove if water is
available

If $P > PE$ → water surplus

If $P < PE$ → water deficit

Get started with the worksheet!

Option to use Excel spreadsheet to graph for question 2

- If you do this option, **upload excel spreadsheet** to the Canvas assignment “Lab 5”
- Make sure to **label axes** and differentiate between P and PE
- **Label** which graph is **Champaign**, and which is **Austin**